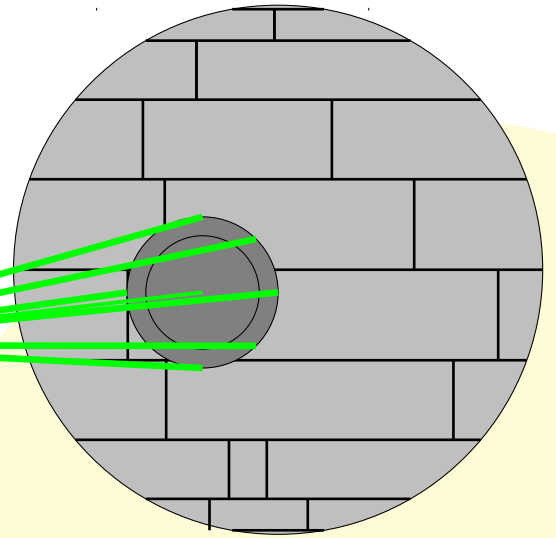


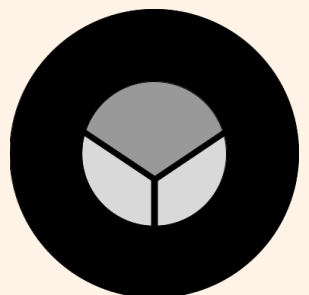
"May the Force Be With You"
— General Jan Dodonna



Q1 Grade Analysis

Daniel Sabalakov

DE Introduction to Probability and Statistics, Portsmouth High School
November 5, 2025



Overview

Grades in Great Bay Stat. are categorized in three ways: Formative, Summative, Final. Each are weighted 20%, 70%, and 10%, respectively; meaning that, technically, only the Summative Grade is required to pass the course. All grades analyzed will not include individual weights.

Let's analyse Daniel Sabalakov's, a student at Great Bay Stat., grades.

Data set

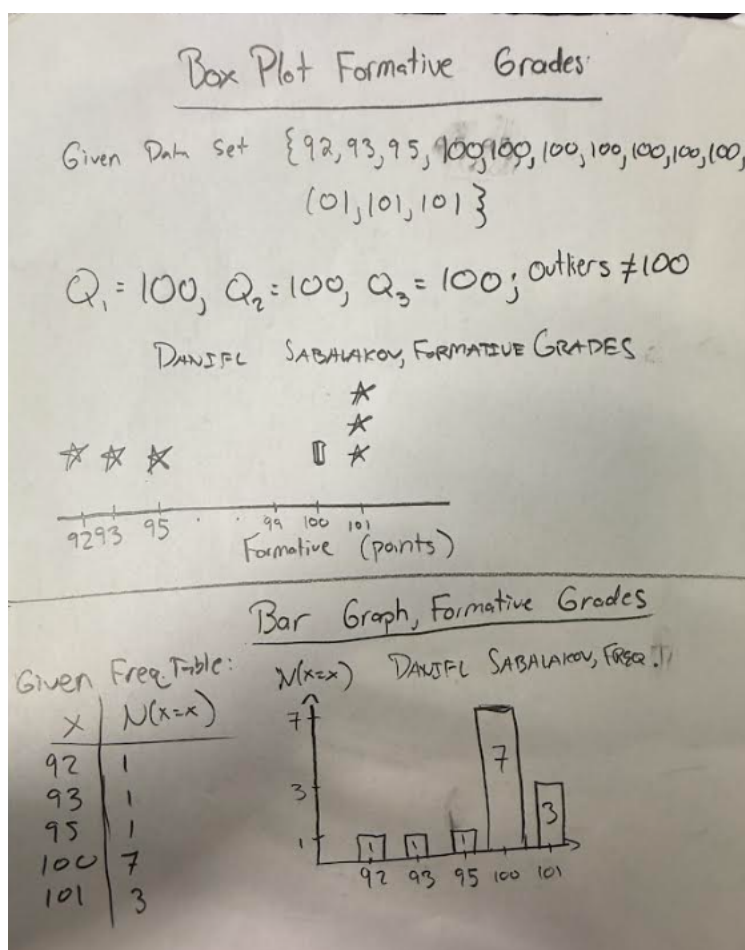
Formative

Frequency Table for Formative:

x	92	93	95	100	101
N(X=x)	1	1	1	7	3

Raw Data for Formative Grades: {100,101,100,100,93,92,101,100,101,100,100,95,100}

Formative Bar Graph and Box Plot



Formative Mean and Standard Deviation, In Context

The mean of the Formative category is 98.692, and the standard deviation is 3.146. The standard deviation of 3.146 points is the typical distance between Daniel Sabalakov's average Formative score of 98.692 points and one of Daniel Sabalakov's scores.

Formative Median and IQR, In Context

The median of the Formative category is 100 points, and the IQR of the Formative category is 0 points. ($Q_3 - Q_1 = 100 - 100 = 0$) This means that the middle point of Daniel Sabalakov's ordered points is a 100 and that the 25th percentile and the 75th percentile are 0 points apart.

Formative Outliers, In Context

Using our two outlier rules of thumb, we can determine the left fence for each outlier.

1. Mean & Standard Deviation R.O.T: *Outliers* $<, > \mu \pm 2\sigma$

- Outliers $< 98.692 - 2 * 3.146 \implies$ Outliers < 92.4
- Outliers $> 98.692 + 2 * 3.146 \implies$ Outliers > 104.984

2. Median & IQR R.O.T: *Outliers* $<, > Q_{1,3} \pm 1.5(Q_3 - Q_1)$

- Outliers $< 100 - 1.5 * (100 - 100) \implies$ Outliers < 100
- Outliers $> 100 + 1.5 * (100 - 100) \implies$ Outliers > 100

Using the Mean & Standard Deviations, scores under 92.4 are considered outliers for the data set. Daniel Sabalakov only had one score that is considered an outlier; the score of 92 is considered an outlier because $92 < 92.4$

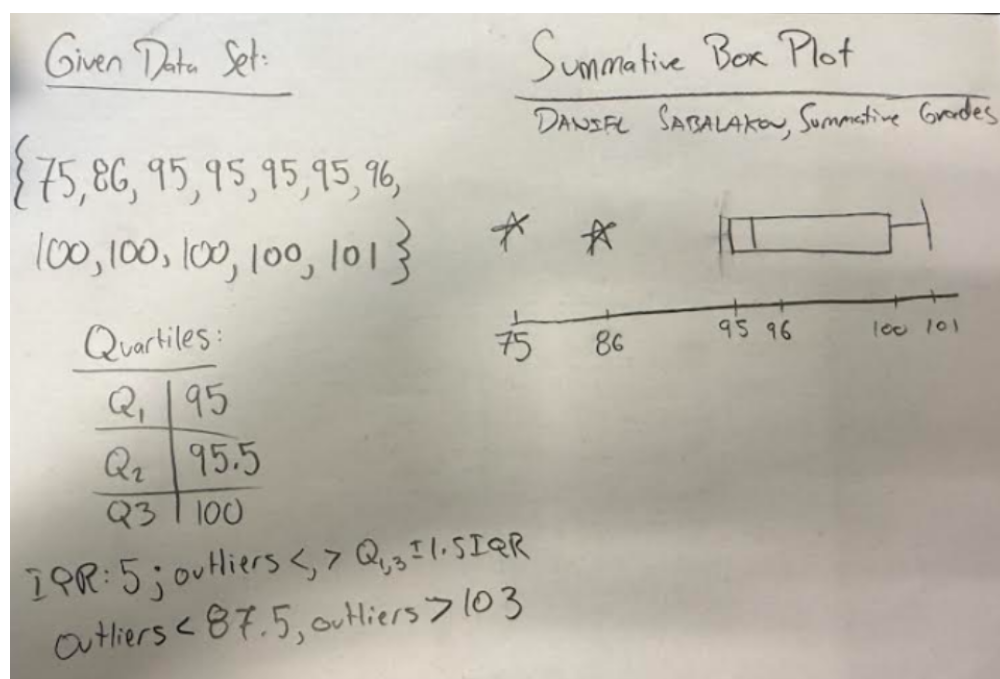
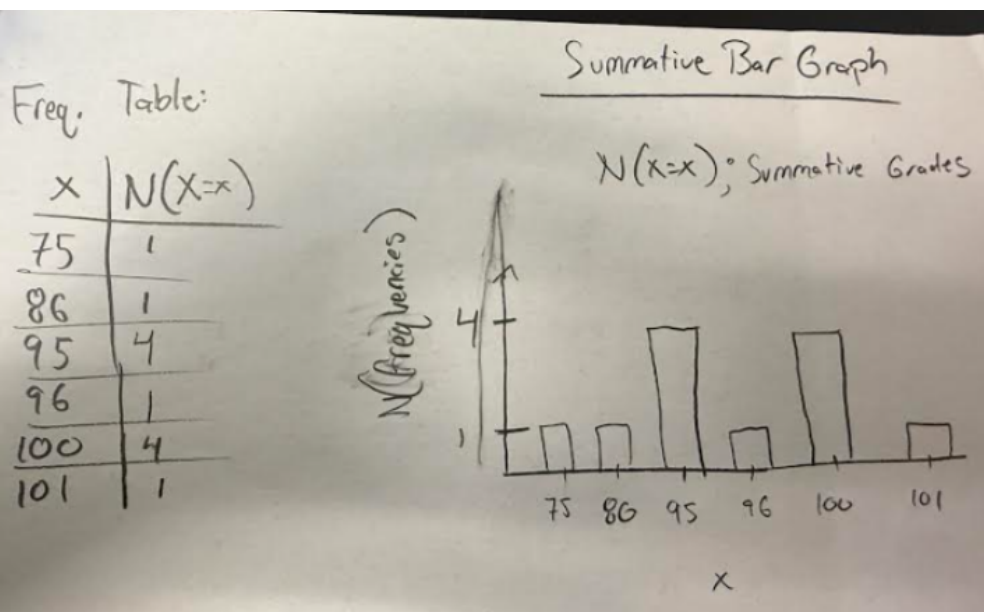
Using the Median & IQR, scores under or over 100 are considered outliers. This yields a result of 6 outlier points. This means that points are very clustered around the 100 point mark since at least 50% of data points in the Formative section are 100's.

Summative

Frequency Table for Summative Grades:

x	75	86	95	96	100	101
N(X=x)	1	1	4	1	4	1

Raw Data for Summative Grades: {96,95,75,100,100,100,101,95,95,86,100,95}



Summative Mean and Standard Deviation, In Context

The mean of the Summative category is 94.833, and the standard deviation is 7.493. The standard deviation of 7.493 points is the typical distance between Daniel Sabalakov's average Summative scores of 94.833 points and one of Daniel Sabalakov's scores.

Summative Median and IQR, In Context

The median of the Summative category is 95.5 points, and the IQR of the Summative category is 5 points ($Q_3 - Q_1 = 100 - 95 = 5$). This means that the middle point of Daniel Sabalakov's ordered points is a 95.5 and that the 25th percentile and the 75th percentile are 5 points apart.

Summative Outliers, In Context

Using our two outlier rules of thumb, we can determine the left fence for each outlier.

1. Mean & Standard Deviation R.O.T: *Outliers* $<, > \mu \pm 2\sigma$

- Outliers $< \$94.833 - 2 \cdot 7.493 \$ \implies$ Outliers < 79.846
- Outliers $> \$94.833 + 2 \cdot 7.493 \$ \implies$ Outliers > 109.819

2. Median & IQR R.O.T: *Outliers* $<, > Q_{1,3} \pm 1.5(Q_3 - Q_1)$

- Outliers $< 95.5 - 1.5 \cdot (5) \implies$ Outliers < 87.5
- Outliers $> 95.5 + 1.5 \cdot (5) \implies$ Outliers > 103

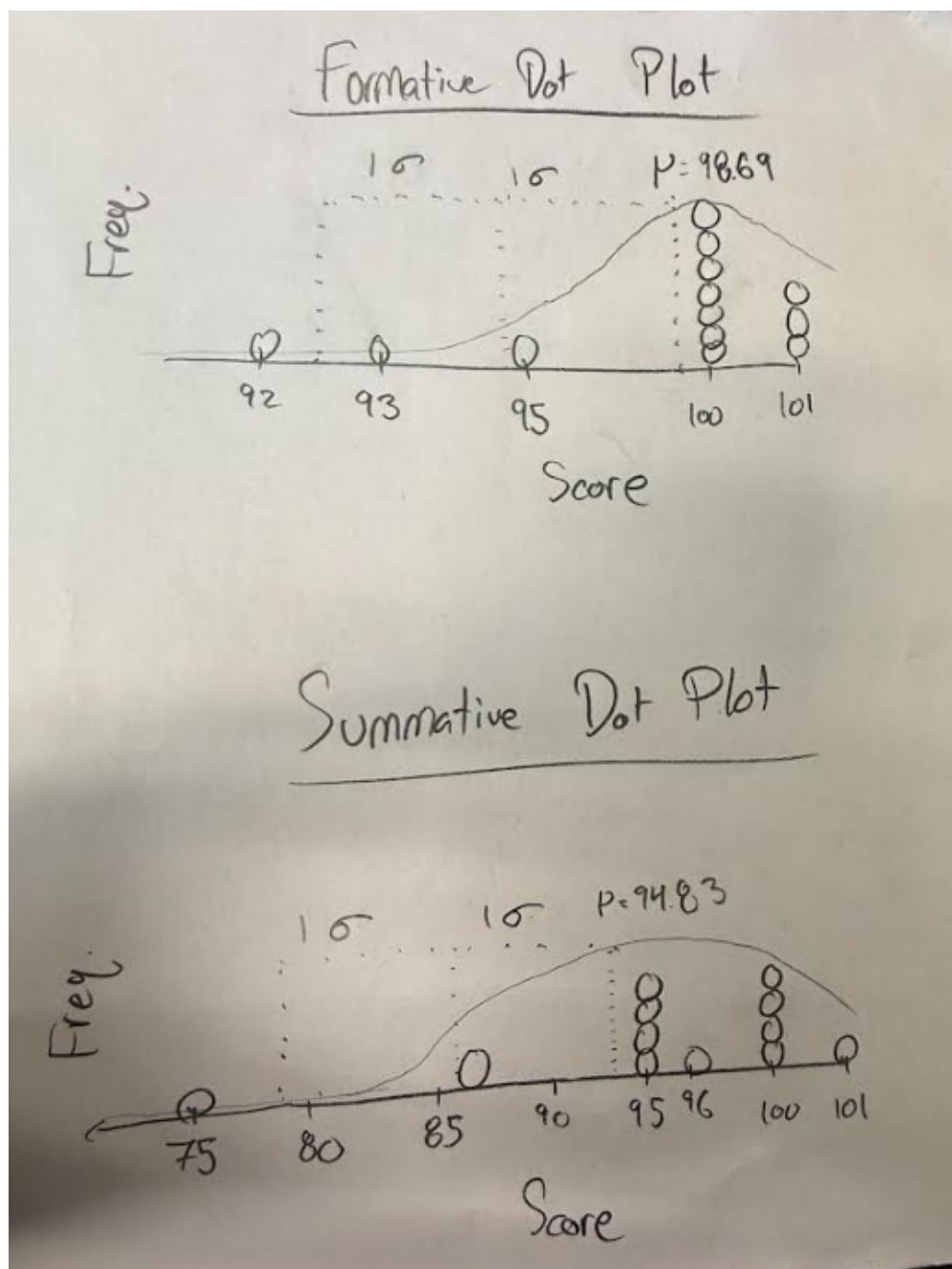
Using the Mean & Standard Deviations, scores under 79.846 are considered outliers for the Summative data set. Daniel Sabalakov only had one score that is considered an outlier; the score of 75 is considered an outlier because $75 < 79.846$.

Using the Median & IQR, scores under a score of 87.5 or over a score of 103 are considered outliers. This yields a result of 2 outlier points: a score of 86 and a score of 75.

Final

Daniel Sabalakov scored exceptionally well on the final, scoring a 100% on each of the three portions of the final. This yields a median and a mean of a 100, and an IQR and Standard Deviation of 0.

Analysis and Conclusion



A score of 75 in the Summative category, quite frankly, is an outlier beyond doubt. Calculating the z-score of a score of 75, we get: $Z = \frac{y - \mu}{\sigma}$, $\Rightarrow Z = \frac{75 - 94.833}{7.493} \Rightarrow Z = -2.669$. Using the Mean and Standard Deviation rule of thumb, any z-score under -2 is considered an outlier. If this is a normal model (which I will discuss in the next section), a Summative score of 75 or lower has a 0.38% chance of occurring. Additionally, using the Median and IQR rule of thumb, any score under 87.5 is an outlier. While Daniel Sabalakov did score a 86, which would be considered an outlier, a score of 75 is even more of an outlier. Hence, a Summative score of 75 is undeniably an outlier from both the Median & IQR perspective and a Mean and Standard Deviation perspective.

For data to be modeled normally, the following characteristics must be true. Of the Summative distribution,

- Symmetry

-
- The Summative distribution is **not** symmetric. An outlier of 75 points as well as an atypical 85 points make the curve left-skewed.
 - Bell Shape (Unimodal):
 - The Summative distribution must be unimodal. The Summative Grade Distribution is **not** unimodal; rather, it is left-skewed.
 - 50%-95%-99% rule
 - The Summative distribution does **not** follow the 50-95-99% rule. A majority of the data is found within 1 standard deviation of the mean; 5/6th of the data is found within 1 standard deviation of the mean. 1/12 of the data (8.3%) is found within 2 standard deviations, and 1/12 of the data (8.3%) is an outlier. (Using the mean & stdev R.O.T)
 - Thin tails
 - The Summative distribution does **not** have thin tails on both asymptotes. On the right, the Summative Distribution is not thin-tailed.

This means that the Summative distribution can not be modeled normally. However, this does not invalidate the point that a score of 75 is an outlier; a normal distribution only contributes a percentage.

Both the Summative and Formative score distributions are Left-skewed, meaning that the mean will be less than the median for both. If the Summative score of 75 (which is clearly an outlier) is removed from the Summative distribution, Daniel's overall grade will be a 96.9%. Removing the Summative score of 75 will additionally change the outlier rule of a typical distribution. The Summative distribution without the 75 has a mean of 96.63 with a standard deviation of 4.34 points. This means that, on average, a score in the modified distribution will vary by 4.34 points from the mean of 96.63 points.

Screenshot of Google Spreadsheet

	A	B	C	D	E	F	G
1	Unsorted				Sorted		
2	Formative	Summative	Final		Formative	Summative	Final
3	100	96	100		101	101	100
4	101	95	100		101	100	100
5	100	75	100		101	100	100
6	100	100			100	100	
7	93	100			100	100	
8	92	100			100	96	
9	101	101			100	95	
10	100	95			100	95	
11	101	95			100	95	
12	100	86			100	95	
13	100	100			95	86	
14	95	95			93	75	
15	100				92		
16							
17	Average				Median		
18	Formative	Summative	Final		Formative	Summative	Final
19	98.69230769	94.83333333	100		100	95.5	100
20	STDEV				IQR		
21	3.146019055	7.493431467	0		0	5	0
22					Q1		
23					100	95	100
24	Outlier Bounds				Outlier Bounds		
25	92.40026958	79.8464704	100		100	87.5	100

29	Without the 75						
30	Unsorted				Sorted		
31	Formative	Summative	Final		Formative	Summative	Final
32	100	96	100		101	101	100
33	101	95	100		101	100	100
34	100	95	100		101	100	100
35	100	100			100	100	
36	93	100			100	100	
37	92	100			100	96	
38	101	101			100	95	
39	100	95			100	95	
40	101	95			100	95	
41	100	86			100	95	
42	100	100			95	86	
43	95				93		
44	100				92		
45							
46	Average				Median		
47	Formative	Summative	Final		Formative	Summative	Final
48	98.69230769	96.63636364	100		100	96	100
49	STDEV				IQR		
50	3.146019055	4.342182107	0		0	5	0
51					Q1		
52					100	95	100
53	Outlier Bounds				Outlier Bounds		
54	92.40026958	87.95199942	100		100	87.5	100
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